LEVEL 2 ASSIGNMENT

STEP PROGRAM

STRINGS

SUBMITTED BY:

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CLASS: AJ1

SOURCE CODE

LEVEL 2 QUESTION: 1

Q. An organization took up the exercise to find the Body Mass Index (BMI) of all the persons in a team of 10 members. For this create a program to find the BMI and display the height, weight, BMI, and status of each individual

```
import java.util.Scanner;


public class BMICalculator {
    public static String[][] computeBMIStatus(double[][] data) {
        String[][] result = new String[10][4];
        for (int i = 0; i < 10; i++) {
            double weight = data[i][0];
            double heightCm = data[i][1];
            double heightM = heightCm / 100.0;
            double bmi = weight / (heightM * heightM);
            String status;
            if (bmi < 18.5)
                status = "Underweight";
            else if (bmi < 24.9)
```

```
            status = "Normal";
        else if (bmi < 29.9)
            status = "Overweight";
        else
            status = "Obese";
        result[i][0] = String.format("%.2f", heightCm);
        result[i][1] = String.format("%.2f", weight);
        result[i][2] = String.format("%.2f", bmi);
        result[i][3] = status;
    }
    return result;
}

public static String[][] processBMI(double[][] hwArray) {
    return computeBMIStatus(hwArray);
}

public static void displayResult(String[][] bmiArray) {
    System.out.println("Height(cm)\tWeight(kg)\tBMI\t\tStatus");
    for (int i = 0; i < 10; i++) {
        System.out.printf("%s\t\t%s\t\t%s\t%s\n",
                bmiArray[i][0], bmiArray[i][1], bmiArray[i][2], bmiArray[i][3]);
    }
}
```

```java
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double[][] hwArray = new double[10][2];
        for (int i = 0; i < 10; i++) {
            System.out.print("Enter weight (kg) for person " + (i + 1) + ": ");
            hwArray[i][0] = sc.nextDouble();
            System.out.print("Enter height (cm) for person " + (i + 1) + ": ");
            hwArray[i][1] = sc.nextDouble();
        }
        String[][] bmiArray = processBMI(hwArray);
        displayResult(bmiArray);
    }
}
```

OUTPUT:

```
Enter your weight in kg: 76
Enter your height in cm: 170
Your BMI is: 26.30
Status: Overweight
(base) harikrishna@Harikrishnas-MacBook-Air-281 java %
```

LEVEL 2 QUESTION: 2

Q. Find unique characters in a string using the charAt() method and display the result

```
import java.util.Scanner;

public class UniqueCharacters {

    public static int findLength(String text) {
        int length = 0;
        try {
            while (true) {
                text.charAt(length);
                length++;
            }
        } catch (IndexOutOfBoundsException e) {
        }
        return length;
    }

    public static char[] findUniqueCharacters(String text) {
        int len = findLength(text);
```

```
char[] unique = new char[len];
int uniqueCount = 0;

for (int i = 0; i < len; i++) {
    char current = text.charAt(i);
    boolean isUnique = true;
    for (int j = 0; j < i; j++) {
        if (current == text.charAt(j)) {
            isUnique = false;
            break;
        }
    }
    if (isUnique) {
        unique[uniqueCount++] = current;
    }
}

char[] result = new char[uniqueCount];
for (int i = 0; i < uniqueCount; i++) {
    result[i] = unique[i];
}
return result;
```

```
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = sc.nextLine();
        char[] uniqueChars = findUniqueCharacters(input);
        System.out.print("Unique characters: ");
        for (char c : uniqueChars) {
            System.out.print(c + " ");
        }
    }
}
```

OUTPUT:

```
(base) harikrishna@Harikrishnas-MacBoo
a
Enter a string: hello
Unique characters: h e l o %
```

LEVEL 2 QUESTION: 3

Q. Write a program to find the first non-repeating character in a string and show the result

```
import java.util.Scanner;

public class FirstNonRepeatingChar {

  public static char findFirstNonRepeating(String text) {
    int[] freq = new int[256];
    int len = 0;

    try {
      while (true) {
        char ch = text.charAt(len);
        freq[ch]++;
        len++;
      }
    } catch (IndexOutOfBoundsException e) {
    }

    for (int i = 0; i < len; i++) {
      char ch = text.charAt(i);
      if (freq[ch] == 1) {
        return ch;
```

```java
            }
        }

        return '\0';
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = sc.nextLine();
        char result = findFirstNonRepeating(input);
        if (result == '\0') {
            System.out.println("No non-repeating character found.");
        } else {
            System.out.println("First non-repeating character: " + result);
        }
    }
}
```

OUTPUT:

```
r.java
Enter a string: world
First non-repeating character: w
```

LEVEL 2 QUESTION: 4

Q. Write a program to find the frequency of characters in a string using the charAt() method and display the result

```
import java.util.Scanner;

public class CharacterFrequency {

  public static String[][] findFrequencies(String text) {
    int[] freq = new int[256];
    int length = 0;

    try {
      while (true) {
        char ch = text.charAt(length);
        freq[ch]++;
        length++;
      }
```

```
} catch (IndexOutOfBoundsException e) {
}

boolean[] counted = new boolean[256];
String[][] result = new String[length][2];
int index = 0;

for (int i = 0; i < length; i++) {
  char ch = text.charAt(i);
  if (!counted[ch]) {
    result[index][0] = String.valueOf(ch);
    result[index][1] = String.valueOf(freq[ch]);
    counted[ch] = true;
    index++;
  }
}

String[][] finalResult = new String[index][2];
for (int i = 0; i < index; i++) {
  finalResult[i] = result[i];
}
```

```
        return finalResult;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = sc.nextLine();
        String[][] frequencies = findFrequencies(input);
        System.out.println("Character\tFrequency");
        for (int i = 0; i < frequencies.length; i++) {
            System.out.println(frequencies[i][0] + "\t\t" + frequencies[i][1]);
        }
    }
}
```

OUTPUT:

```
Enter a string: car
Character       Frequency
c               1
a               1
r               1
```

LEVEL 2 QUESTION: 5

Q. Write a program to find the frequency of characters in a string using unique characters and display the result

```
import java.util.Scanner;

public class CharacterFrequencyWithUnique {

   public static char[] uniqueCharacters(String text) {
      int length = 0;
      try {
         while (true) {
            text.charAt(length);
            length++;
         }
      } catch (IndexOutOfBoundsException e) {
      }

      char[] unique = new char[length];
      int count = 0;

      for (int i = 0; i < length; i++) {
         char current = text.charAt(i);
```

```
        boolean isUnique = true;
        for (int j = 0; j < i; j++) {
            if (current == text.charAt(j)) {
                isUnique = false;
                break;
            }
        }
        if (isUnique) {
            unique[count++] = current;
        }
    }

    char[] result = new char[count];
    for (int i = 0; i < count; i++) {
        result[i] = unique[i];
    }
    return result;
}

public static String[][] findFrequencies(String text) {
    int[] freq = new int[256];
    int len = 0;
```

```
    try {
        while (true) {
            char ch = text.charAt(len);
            freq[ch]++;
            len++;
        }
    } catch (IndexOutOfBoundsException e) {
    }

    char[] uniqueChars = uniqueCharacters(text);
    String[][] result = new String[uniqueChars.length][2];

    for (int i = 0; i < uniqueChars.length; i++) {
        result[i][0] = String.valueOf(uniqueChars[i]);
        result[i][1] = String.valueOf(freq[uniqueChars[i]]);
    }

    return result;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a string: ");

        String input = sc.nextLine();

        String[][] frequencies = findFrequencies(input);

        System.out.println("Character\tFrequency");

        for (int i = 0; i < frequencies.length; i++) {

            System.out.println(frequencies[i][0] + "\t\t" + frequencies[i][1]);

        }

    }

}
```

OUTPUT:

```
Enter a string: hello
Character       Frequency
h               1
e               1
l               2
o               1
(base) harikrishna@Harikrish
```

LEVEL 2 QUESTION: 6

Q. Write a program to find the frequency of characters in a string using nested loops and display the result

```java
import java.util.Scanner;

public class FrequencyWithNestedLoops {

    public static String[] findFrequencies(String text) {
        char[] chars = text.toCharArray();
        int[] freq = new int[chars.length];

        for (int i = 0; i < chars.length; i++) {
            freq[i] = 1;
            if (chars[i] == '0') {
                continue;
            }
            for (int j = i + 1; j < chars.length; j++) {
                if (chars[i] == chars[j]) {
                    freq[i]++;
                    chars[j] = '0';
                }
            }
        }
```

```
        int count = 0;
        for (int i = 0; i < chars.length; i++) {
            if (chars[i] != '0') {
                count++;
            }
        }

        String[] result = new String[count];
        int index = 0;
        for (int i = 0; i < chars.length; i++) {
            if (chars[i] != '0') {
                result[index] = chars[i] + " = " + freq[i];
                index++;
            }
        }

        return result;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
```

```
        String input = sc.nextLine();

        String[] frequencies = findFrequencies(input);

        System.out.println("Character Frequencies:");

        for (String entry : frequencies) {

            System.out.println(entry);

        }

    }

}
```

OUTPUT:

```
Enter a string: world
Character Frequencies:
w = 1
o = 1
r = 1
l = 1
d = 1
```

LEVEL 2 QUESTION: 7

Q. Write a program to to check if a text is palindrome and display the result

```
import java.util.Scanner;


public class PalindromeCheck {
```

```
public static boolean isPalindromeIterative(String text) {
    int start = 0;
    int end = getLength(text) - 1;

    while (start < end) {
        if (text.charAt(start) != text.charAt(end)) {
            return false;
        }
        start++;
        end--;
    }
    return true;
}

public static boolean isPalindromeRecursive(String text, int start, int end) {
    if (start >= end) {
        return true;
    }
    if (text.charAt(start) != text.charAt(end)) {
        return false;
    }
    return isPalindromeRecursive(text, start + 1, end - 1);
```

```
}

public static boolean isPalindromeArrayMethod(String text) {
    char[] original = text.toCharArray();
    char[] reversed = reverseString(text);
    for (int i = 0; i < original.length; i++) {
        if (original[i] != reversed[i]) {
            return false;
        }
    }
    return true;
}

public static char[] reverseString(String text) {
    int len = getLength(text);
    char[] reversed = new char[len];
    for (int i = len - 1, j = 0; i >= 0; i--, j++) {
        reversed[j] = text.charAt(i);
    }
    return reversed;
}
```

```java
public static int getLength(String text) {
    int len = 0;
    try {
        while (true) {
            text.charAt(len);
            len++;
        }
    } catch (IndexOutOfBoundsException e) {
    }
    return len;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a string: ");
    String input = sc.nextLine();

    boolean result1 = isPalindromeIterative(input);
    boolean result2 = isPalindromeRecursive(input, 0, getLength(input) - 1);
    boolean result3 = isPalindromeArrayMethod(input);

    System.out.println("Palindrome check using Iterative logic: " + result1);
```

```
        System.out.println("Palindrome check using Recursive logic: " + result2);

        System.out.println("Palindrome check using Array logic: " + result3);

    }

}
```

OUTPUT:

```
Enter a string: bike
Palindrome check using Iterative logic: false
Palindrome check using Recursive logic: false
Palindrome check using Array logic: false
(base) harikrishna@Harikrishnas-MacBook-Air-281 java % x
```

LEVEL 2 QUESTION: 8

Q. Write a program to check if two texts are anagrams and display the result

```
import java.util.Scanner;

public class AnagramCheck {

    public static boolean areAnagrams(String text1, String text2) {

        int len1 = getLength(text1);
```

```
int len2 = getLength(text2);

if (len1 != len2) {
    return false;
}

int[] freq1 = new int[256];
int[] freq2 = new int[256];

for (int i = 0; i < len1; i++) {
    freq1[text1.charAt(i)]++;
    freq2[text2.charAt(i)]++;
}

for (int i = 0; i < 256; i++) {
    if (freq1[i] != freq2[i]) {
        return false;
    }
}
```

```
    return true;
}

public static int getLength(String text) {
    int len = 0;
    try {
        while (true) {
            text.charAt(len);
            len++;
        }
    } catch (IndexOutOfBoundsException e) {
    }
    return len;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter first text: ");
```

```
        String input1 = sc.nextLine();

        System.out.print("Enter second text: ");

        String input2 = sc.nextLine();


        boolean result = areAnagrams(input1, input2);

        if (result) {

            System.out.println("The texts are anagrams.");

        } else {

            System.out.println("The texts are not anagrams.");

        }

    }

}
```

OUTPUT:

```
(base) harikrishna@Harikrishnas-Ma
Enter first text: hello
Enter second text: world
The texts are not anagrams.
```

LEVEL 2 QUESTION: 9

Q. Create a program to display a calendar for a given month and year. The program should take the month and year as input from the user and display the calendar for that month. E.g. for 07 2005 user input, the program should display the calendar as shown below

```
import java.util.Scanner;

public class CalendarDisplay {

   public static String getMonthName(int month) {
      String[] months = {
         "", "January", "February", "March", "April", "May", "June",
         "July", "August", "September", "October", "November", "December"
      };
      return months[month];
   }

   public static boolean isLeapYear(int year) {
      return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
```

```
    }

    public static int getNumberOfDaysInMonth(int month,
int year) {
        int[] days = {
            0, 31, 28, 31, 30, 31, 30,
            31, 31, 30, 31, 30, 31
        };
        if (month == 2 && isLeapYear(year)) {
            return 29;
        }
        return days[month];
    }

    public static int getFirstDayOfMonth(int day, int month,
int year) {
        int y0 = year - (14 - month) / 12;
        int x = y0 + y0 / 4 - y0 / 100 + y0 / 400;
        int m0 = month + 12 * ((14 - month) / 12) - 2;
```

```
        int d0 = (day + x + (31 * m0) / 12) % 7;
        return d0;
    }

    public static void displayCalendar(int month, int year) {
        String monthName = getMonthName(month);
        int daysInMonth = getNumberOfDaysInMonth(month, year);
        int startDay = getFirstDayOfMonth(1, month, year);

        System.out.printf("    %s %d\n", monthName, year);
        System.out.println("Sun Mon Tue Wed Thu Fri Sat");

        for (int i = 0; i < startDay; i++) {
            System.out.print("    ");
        }

        for (int day = 1; day <= daysInMonth; day++) {
            System.out.printf("%3d ", day);
```

```
            if ((day + startDay) % 7 == 0) {
                System.out.println();
            }
        }
        System.out.println();
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter month (1-12): ");
        int month = sc.nextInt();
        System.out.print("Enter year: ");
        int year = sc.nextInt();

        displayCalendar(month, year);
    }
}
```

OUTPUT:

```
Enter month (1-12): 12
Enter year: 2005
     December 2005
Sun Mon Tue Wed Thu Fri Sat
                  1   2   3
  4   5   6   7   8   9  10
 11  12  13  14  15  16  17
 18  19  20  21  22  23  24
 25  26  27  28  29  30  31
```

LEVEL 2 QUESTION: 10

Q. Write a program to create a deck of cards, initialize the deck, shuffle the deck, and distribute the deck of n cards to x number of players. Finally, print the cards the players have

```
import java.util.Scanner;

public class DeckOfCards {

    public static String[] initializeDeck() {
        String[] suits = { "Hearts", "Diamonds", "Clubs", "Spades" };
        String[] ranks = { "2", "3", "4", "5", "6", "7", "8", "9", "10", "Jack",
"Queen", "King", "Ace" };
        int numOfCards = suits.length * ranks.length;
        String[] deck = new String[numOfCards];
```

```
    int index = 0;
    for (String suit : suits) {
        for (String rank : ranks) {
            deck[index++] = rank + " of " + suit;
        }
    }
    return deck;
}

public static String[] shuffleDeck(String[] deck) {
    int n = deck.length;
    for (int i = 0; i < n; i++) {
        int randomCardNumber = i + (int) (Math.random() * (n - i));
        String temp = deck[i];
        deck[i] = deck[randomCardNumber];
        deck[randomCardNumber] = temp;
    }
    return deck;
```

```
    }

    public static String[][] distributeCards(String[] deck, int n, int x) {
        if (x <= 0 || n <= 0 || n > deck.length || n % x != 0) {
            System.out.println("Invalid distribution. Ensure n <= 52 and n divisible by x.");
            return null;
        }
        int cardsPerPlayer = n / x;
        String[][] players = new String[x][cardsPerPlayer];
        int index = 0;
        for (int i = 0; i < x; i++) {
            for (int j = 0; j < cardsPerPlayer; j++) {
                players[i][j] = deck[index++];
            }
        }
        return players;
    }
```

```
public static void printPlayerCards(String[][] players) {
    if (players == null) return;
    for (int i = 0; i < players.length; i++) {
        System.out.println("Player " + (i + 1) + ":");
        for (int j = 0; j < players[i].length; j++) {
            System.out.println("  " + players[i][j]);
        }
        System.out.println();
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number of cards to distribute (<=
52): ");
    int n = sc.nextInt();
    System.out.print("Enter number of players: ");
    int x = sc.nextInt();
```

```java
        String[] deck = initializeDeck();
        deck = shuffleDeck(deck);
        String[][] players = distributeCards(deck, n, x);
        printPlayerCards(players);
    }
}
```

OUTPUT:

```
Enter number of cards to distribute (<= 52): 10
Enter number of players: 5
Player 1:
  Queen of Hearts
  King of Diamonds

Player 2:
  Jack of Diamonds
  2 of Clubs

Player 3:
  9 of Hearts
  2 of Spades

Player 4:
  5 of Diamonds
  10 of Clubs

Player 5:
  6 of Hearts
```